geoagent

An AI agent that works out where a 360° panorama was taken, and records how it got there.



One of our evaluation panoramas, on the Madeira plateau (Mapillary, © geouma, CC BY-SA). The agent placed it 9 m from the true position: see page 4.

THE TASK

One 360° panorama, its capture time and an **Urban** or **non-Urban** label. Mostly Portugal, also Spain and France. Street names are rare and some panoramas are in open fields.

WHAT IT RETURNS

Ten ranked (lat, lng) guesses per panorama, and a JSON log of every interaction: each model turn, tool call and human input.

THE GOAL

Solve the scoring run **on its own**. People may help only by answering the agent's question, writing a cue or placing a pin, and each of those costs points.

HOW IT WORKS

The model proposes weighted hypotheses. Code builds the answer.

INPUT
Panorama
6080 × 3040 px image, timestamp, Urban / non-Urban label.

→

PREPARE
11 views
An overview with a compass ruler, 8 perspective views, one up, one down. Sent whole, signs would be unreadable.

→

REASON
Agent loop
`gpt-6.1-sol` at high reasoning effort looks, calls tools and revises. Up to 30 turns and 14 minutes; 45 and 20 for the Special One.

→

PROPOSE
Hypotheses
Each one a point, a probability, an uncertainty radius and its evidence.

→

DECIDE
Allocator
Picks the point with the highest expected score and fills the 10 slots.

→

OUTPUT
Answer + log
`answers.csv` and one JSON log per panorama.

TOOLS THE AGENT CAN CALL

`view_crop`
Zoom into the panorama: signs, plates, shopfronts, distant landmarks.

`search_places`
Turn a name read off a sign into coordinates (Nominatim, Photon, Google).

`web_search`
Look up business names, phone numbers, landmarks.

`sun_position`
Sun direction plus timestamp gives the compass heading of the panorama.

`streetview_lookup`
Street-level photos near a candidate (Google Street View, Mapillary) to compare.

`satellite_tile`
Aerial image: road layout, roofs, field patterns, coastline.

`overpass_query`
Query OpenStreetMap for feature combinations: a church near a roundabout.

`ask_multiple_choice`
One question to the operator, at most 4 options. Costs 5% of the score.

HOW IT REASONS: COARSE TO FINE

1. **Survey**
   List the evidence before naming any place.
2. **Hypothesise**
   Country, region, municipality, with probabilities.
3. **Read text**
   Zoom on anything legible and look it up.
4. **Orient**
   Use the sun to restate what it sees in compass terms.
5. **Generate candidates**
   From geocoded text or map feature queries.
6. **Verify**
   Street-level views for Urban, satellite for non-Urban.
7. **Decide on a question**
   Ask only if the answer is worth its cost.
8. **Submit**
   Hypotheses with probabilities and radii.

> **Taught up front, where it is free:** the system prompt carries a regional playbook for Portugal, Spain and France (sign vocabulary, postcode and phone formats, road numbering, bollards, architecture, vegetation) and the scoring rules, so the agent knows what precision is worth.

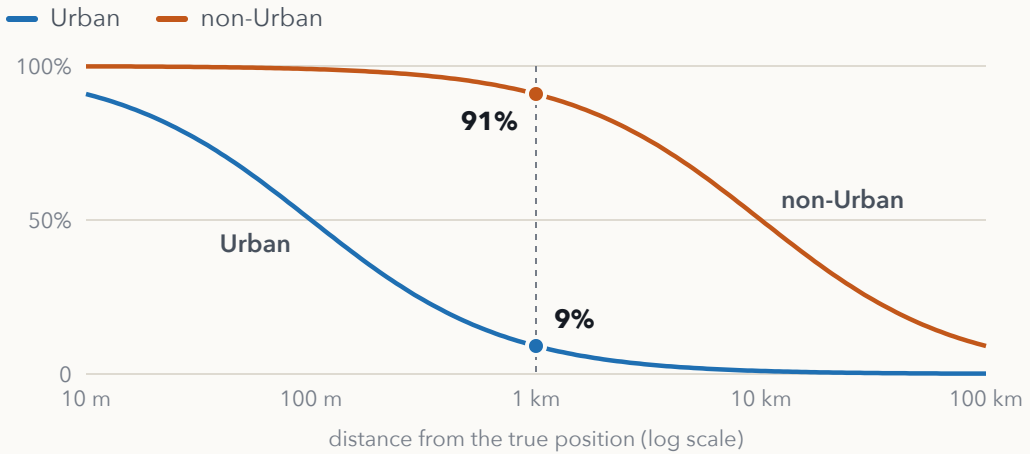
STRATEGY

The scoring rules decide how the agent answers.

```
score = Σ  1000 × 0.95^(rank−1) × θ(distance) × penalties
θ Urban     = 2 / (1 + d / 100 m)
θ non−Urban = 2 / (1 + d / 10 km)
```

Under 10 m counts as exact. At most 16,050 points per panorama; the Special One counts three times.

Share of the maximum score kept, by distance from the true position



1 Always ten entries
A single entry earns 12.5% of what ten copies of it earn.

2 The best point, ten times
Expected score is linear in the entries, so every rank is best filled by the same point. Spreading slots over several guesses lowers it.

3 Never the midpoint
The score curve is convex, so the likelier of two places always beats the point between them.

4 Urban is street matching; non-Urban is region finding
One kilometre off keeps 9% of an Urban score and 91% of a non-Urban one. Precision effort goes to Urban panoramas and the Special One.

5 Pin the camera, not the landmark
In a town, a pin 60 m from where the camera stood loses 37%.

HUMAN HELP	FACTOR	WORTH IT ONLY IF THE EXPECTED SCORE RISES BY MORE THAN	HOW THE HARNESS ENFORCES THE RULE
Multiple-choice answer	**× 0.95**	5.3%	Buttons appear only when the agent asks. No text field, at most 3 questions per panorama.
Written cue	**× 0.80**	25%	1000 characters. A lint blocks coordinates, place names, postcodes and road numbers before sending.
Human places the pin	**× 0.50**	100%	Only for a place the agent has named. Ends that panorama, so nothing else can be charged.

IN PRACTICE

Built for a 60-minute scoring run, six panoramas at once.

IT ALWAYS ANSWERS

A provisional 10-entry list is saved every time the hypotheses change. At the soft deadline the agent is told to commit; at the hard deadline, or after a crash or an API failure, **the harness submits the latest list**. With no hypothesis at all it falls back to a prior point for the label.

OPERATOR CONSOLE

One command opens a browser console: a preflight table of the parsed labels, then **one live tile per panorama** with the map, the reasoning stream and the current multiplier. The three penalised controls are the only inputs that reach the agent, and each shows its cost before it is charged.

EVERYTHING IS LOGGED

An append-only JSON log per panorama, flushed per event: model requests and responses with reasoning summaries, tool calls, human inputs, candidate updates and the final list with the multiplier on each entry. Exports can be rebuilt from the logs after a hard kill.

CHECKED OFFLINE

The scoring formula, the penalty ledger, the allocator and the console are covered by **1,033 passing tests** that need no API key or network. True locations are never disclosed, so we score against our own ground-truth set built from Mapillary 360° imagery.

ONE REAL RUN, NO HUMAN HELP

The panorama from page 1, labelled non-Urban.

- `0:16` First hypotheses from vegetation, mist and timber road railings: **Paul da Serra, Madeira, 60%**, within 6.5 km. Runner-up: the uplands of São Miguel, Azores.
- `2:01` Landscape matches on street-level imagery; no road matches yet. 72%.
- `3:22` Still no matching road. Confidence **drops to 55%** and it checks a second plateau.
- `4:29` Street-level imagery near the Estanquinhos campsite matches: **97%, within 500 m**.
- `5:55` Road layout verified at a junction: **98.5%, within 80 m**.
- `6:19` Submitted: the same point in all ten slots.

9 m	20	6:19	$0.62
from the true position (counts as exact)	model turns, 67 tool calls	minutes, of 14 allowed	estimated API cost

One run, not a benchmark. Our ground truth comes from Mapillary, and the agent found Mapillary imagery taken the same day nearby, an advantage the organisers' panoramas may not offer.